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Left-Right Symmetry Breaking in CLIP-style VLMs

A controllable 1D testbed for how relational (left/right) competence actually forms — and a mechanistic answer

> **Central claim** — CLIP-style contrastive training on a 1D image-text testbed *does* learn left-right relations, and an **attention decomposition** shows the mechanism: positional × token embedding interactions induce a **horizontal attention gradient** that breaks left-right symmetry. Ablate it, and discrimination collapses.

Takaki Yamamoto · Chihiro Noguchi · Toshihiro Tanizawa — `arXiv:2601.12809 (v2)` · ICML 2026

1D testbed
controllable image-text scenes, paired left/right descriptions

Label > layout
label diversity is the primary driver of generalization

Attention
decomposition finds the symmetry-breaking gradient

Ablation
removing the term substantially cuts left-right discrimination

Question is spatial understanding truly acquired? › **Testbed** synthetic 1D scenes · CLIP objective › **Mechanism** positional×token → horizontal attention gradient

01 · PROBLEM Does a VLM really know "left of"?

Spatial understanding remains a key challenge in vision-language models — and it is unclear whether apparent competence is truly acquired, and through **what mechanism**. Known symptoms in CLIP-style models include weak parity between mirrored relations.

Instead of benchmarking a giant model, this paper builds the smallest **controllable** setting in which relational competence demonstrably forms — so the emergence can be studied, not just scored.

02 · SETUP Controllable 1D image-text testbed

1D scenes 1–2 objects › **paired left/right descriptions** › **CLIP-style contrastive training** › **unseen object pairs test**

Lightweight Transformer-based **vision and text encoders** are trained end-to-end on one- and two-object scenes, then evaluated on **generalization to unseen object pairs** while varying **label diversity** and **layout diversity** systematically.

03 · HOW TO READ A toy testbed with a scope

Everything is deliberately minimal: **1D scenes with one or two objects**, lightweight Transformer encoders, CLIP-style contrastive loss. "Generalization" means unseen object pairs inside this testbed — that controlled scale is what makes mechanism claims possible at all.

04 · FINDING A Contrastive training learns left-right — conditionally

- Contrastive training **does** acquire left-right relational competence in this testbed.
- **Label diversity matters more than layout diversity** — the primary driver of generalization to unseen pairs.
- So the "does it generalize?" answer depends on how varied the *descriptions* are, not merely how the scene is arranged.

> **Data design implication:** to teach relational grounding, invest in describing the same relation in many ways — mirroring augmentation alone is not the lever.

05 · READING NOTES Three levels of generalization

The testbed scores **three distinct generalization types** (Figs. 3·9·11): **single-object positional** (does an object's position map to "left"/"right" text), **seen-pair** (relations for object pairs observed in training), and **unseen-pair** (compositional transfer to novel pairs). Training dynamics (Fig. 11) track when each level switches on.

Why "symmetry breaking" is the right frame: mirror-symmetric inputs yield symmetric representations unless something *breaks* the symmetry — here the break is shown to be created by the model's own positional interaction, a mechanistic complement to dataset-level bias accounts.

06 · FINDING B Symmetry is broken inside attention

Decomposing the pre-softmax attention logit (Figs. 5·12–14) isolates which term carries the left/right signal: interactions between **positional and token embeddings** induce a **horizontal attention gradient** — a stable left/right asymmetry in how heads attend along the scene axis. The reduced 1-layer model (Fig. 4) isolates the effect with a minimal parameter budget.

Position × token
interaction → horizontal gradient (Fig. 5·14)

Ablated
removing it substantially cuts left–right discrimination

Control
non-generalizing models lack the gradient (Fig. 15)

07 · WHAT TUNES IT Levers on the breaking

- **Weight decay** — Fig. 10 scans it and shows unseen-pair generalization depends on the value: the regularization level shapes whether the symmetry-breaking solution survives.
- **Label vs layout diversity** — systematic variation shows label diversity is the primary driver (Figs. 3·8).
- **Embedding geometry** — Fig. 7 probes image/text token alignment via learned rotation matrices, connecting the gradient to representational space.

> **Mechanism check:** the strongest evidence pairing (Fig. 14 present & generalizing vs Fig. 15 absent & non-generalizing) shows the gradient *tracks* competence — correlation aligned with the ablation's causal cut.

08 · CONFIG & PROTOCOL How the toy is built — Figs 4·10

- **Fig 4 —** reduced 1-layer model: **MB = Mrep = 1, Mh = 4** — the minimal setting where the symmetry either breaks on its own or does not.
- **Fig 10 —** generalizing config: **MB = 2, Mrep = 2, Mh = 4, dhead = 32** — the parameterization behind the mechanism scans.
- **Figs 12·13 —** weight/bias decomposition of the breaking pathway; **Fig 15** is the non-generalizing control that lacks it.

09 · OPEN LIMITS What remains unresolved?

Transfer of the mechanism to full-resolution, multi-object scenes and to production VLMs is beyond the testbed's evidence; the paper's own discussion frames the boundary. Production-scale validation would repeat the decomposition pipeline on trained CLIP checkpoints.

10 · SESSION Team poster #4 of 4

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Reference

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